GRC Risk Treatment & Control Verification Evidence Package

Student / Submitter: GRC Security Engineering Student

Target Host: Ubuntu Server (`192.168.64.11`)

Testing Host: Kali Linux (`192.168.64.7`)

Course Basis: GRC103 Module 3 / ICDFA Risk Treatment Strategies

Date: November 8, 2026

Executive Summary & Submission Overview

This submission ZIP package contains all reproducible terminal outputs, configuration files, redacted log extracts, verification screenshots, and calculation workbooks supporting the **GRC Technical Risk Treatment Report** (`grc_treatment_report.pdf`).

Every requirement from the assignment marking criteria is mapped directly to a specific artifact contained within this archive below.

Evidence Mapping to Rubric & Steps

1. Baseline Environment & Initial Risk Register (Rubric: 15 Marks)

- **Step 1 (Scope & Setup):** `outputs/baseline-ports.txt` and `screenshots/01_ubuntu_ip_address.png` confirm network layout (`192.168.64.11` Target, `192.168.64.7` Kali).
- **Step 4 (Baseline Evidence):** `outputs/baseline-web.txt`, `outputs/baseline-listing.txt`, `outputs/baseline-file.txt`, and `outputs/baseline-legacy.txt` confirm initial active exposures (R1–R4).
- **Report Reference:** Section 2, Table 1 (Initial Risk Register).

2. Avoidance & Continuity Baseline Tests (Rubric: 10 Marks)

- **Step 5 & 6 (Avoidance Execution):** `outputs/avoidance-ports.txt` and `outputs/avoidance-legacy.txt` prove port 8080 is `closed` while port 80 remains `200 OK`.
- **Screenshots:** `screenshots/04_avoidance_verification.png`.
- **Report Reference:** Section 3 (Analytical rationale explaining why disabling 8080 avoids R1 exposure but leaves port 80 risks unmitigated).

3. Archive Hardening & Firewall Validation (Rubric: 25 Marks)

- **Step 7 (Archive Relocation & Path Block):** `configs/nginx_grc-lab.conf` exhibits `autoindex off;` and `location /archive/ { deny all; }`. `screenshots/05_archive_hardening_403.png` confirms `HTTP 403 Forbidden` for direct-file requests.
- **Step 8 (UFW Boundary Control):** `configs/ufw_rules.txt` and `screenshots/06_ufw_status_numbered.png` prove rule enforcement. Blocked-source test (`192.168.64.99`) connection timeout documented in `logs/ufw_kernel.log`.
- **Report Reference:** Section 4 (Defense-in-depth explanation for moving archive out of web root).

4. Log Correlation (Rubric: 10 Marks)

- **Step 9 (Correlate Control Evidence):** `logs/grc_access.log` records HTTP 200 (`/public/`) and HTTP 403 (`/archive/`) at `08/Nov/2026:11:44:22`. `logs/ufw_kernel.log` records dropped packets from unapproved source (`192.168.64.99`) at `Nov 08 11:33:53`.
- **Screenshots:** `screenshots/07_log_correlation.png`.
- **Report Reference:** Section 5, Table 2 (Evidence Correlation Table & Architectural Layer Explanation).

5. Backup & Verified Restoration (Rubric: 15 Marks)

- **Step 10 (Disaster Recovery):**
  - Start Timestamp: `2026-11-08T12:05:24+00:00` (`outputs/portal-before.sha256`)
  - Completion Timestamp: `2026-11-08T12:09:34+00:00`
  - Measured RTO: **4 minutes 10 seconds**
  - Verification: `screenshots/10_hash_verification_rto.png` showing `index.html: OK`.
- **Report Reference:** Section 6 (Analysis of same-host backup limitations).

6. Treatment Governance & Financial Reasoning (Rubric: 15 Marks)

- **Step 11 (Transfer & Acceptance):** Managed backup provider clause draft and simulated 14-day risk acceptance approval record (expiring Nov 22, 2026).
- **Step 12 (Cost Modeling & ALE):** `calculations/risk_cost_model.csv` details Options A/B/C first-year costs ($150, $360, $100). $ALE financial calculations show Baseline ALE$(1,000), Residual ALE ($200), and $Net First - Year Benefit$(440). Post-treatment reassessment of R1–R4.
- **Report Reference:** Sections 7, 8, and 9.

7. Reproducible Evidence & Management Report (Rubric: 10 Marks)

- **Deliverable:** `grc_treatment_report.pdf` (Self-contained, 3-page management PDF).

Reproducibility Commands Summary

To verify the evidence package integrity from a clean terminal:

```
None
# 1. Verify SHA-256 hash of original portal backup
sha256sum -c outputs/portal-before.sha256

# 2. Inspect Nginx hardened location block
cat configs/nginx_grc-lab.conf | grep -A 3 "location /archive/"

# 3. Verify firewall default incoming policy
cat configs/ufw_rules.txt | grep "Default:"

# 4. Inspect financial model spreadsheet
cat calculations/risk_cost_model.csv
```